MTH2003 Examples Sheet 1

Learn how to solve first order ODEs by practising all of these examples. Feel free to work on these examples with your friends and ask me about any queries during the drop-in Q/A sessions on Mondays at 16:35 (see your timetable).

1. For each of the differential equations below, determine whether they can be solved:

   [i] by the method of separation of variables,
   [ii] as a homogeneous first-order ODE,
   [iii] as a linear first-order ODE using an integrating factor.

   For each method that is applicable, find the solutions and identify the intervals on which your solutions are defined.

   (a) $\dfrac{\mathrm{d}y}{\mathrm{d}t} = 2 - 3y.$
   (b) $\dfrac{1}{y}\dfrac{\mathrm{d}y}{\mathrm{d}t} - \dfrac{1}{t} = 0.$
   (c) $\dfrac{\mathrm{d}y}{\mathrm{d}t} = t^3 - y^3.$

2. For each of the following ODEs:

   [i] Analyse the right-hand side and determine for what pairs (t_0, y_0) the initial value problem $y(t_0) = y_0$ is guaranteed to have locally

   - a solution by Peano's theorem
   - a unique solution by Picard's theorem.

   [ii] Use the method of separation of variables to find the special solutions and typical solutions. For each solution, identify the maximal interval in which it is defined, and the maximal interval in which it is unique. Determine whether the two classes (i.e. the special solutions and the typical solutions) can be combined into a single formula.
   [iii] Sketch a graph showing a representative set of typical solutions and any special solutions. Use this diagram to identify possible points of uniqueness violation. Compare these results with your prediction in part [i].
   [iv] Find the solutions satisfying the specified initial conditions, and determine the maximal intervals where they exist, and maximal intervals where they are unique. Sketch the graphs of these solutions, indicating the initial points.

   (a) $y' = ty$, (A) $y(1) = 1$, (B) $y(1) = 0$, (C) $y(0) = 1$, (D) $y(0) = 0$.
   (b) $y' = 2ty^2$, (A) $y(1) = 0$, (B) $y(0) = 1$.